

Multiple Linear Regression

From One Predictor to Many

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Fall 2026

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Roadmap

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1. From Simple to Multiple Regression

1.1 Why More Than One Predictor

Real outcomes are associated with many inputs at once:

- Unit sales are associated with TV *and* radio *and* newspaper ad spend simultaneously.
- Starting salary is associated with GPA, internship experience, major, school.
- House price is associated with square footage, location, age, bedrooms.

Running a separate simple linear regression for each predictor (one-at-a-time) cannot answer the natural question:

“What is the conditional association between X_j and Y , holding the other predictors fixed?”

Multiple regression handles all predictors at once and gives exactly that conditional answer.

1.2 The Model

Definition — multiple linear regression

With p predictors,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon,$$

where $\mathbb{E}[\epsilon \mid X_1, \dots, X_p] = 0$ and $\text{Var}(\epsilon \mid \mathbf{X}) = \sigma^2$ (homoscedasticity).

Across the sample, the observed pairs $(\mathbf{X}_i, Y_i)$ are independent across i , following the same convention as L6–L7.

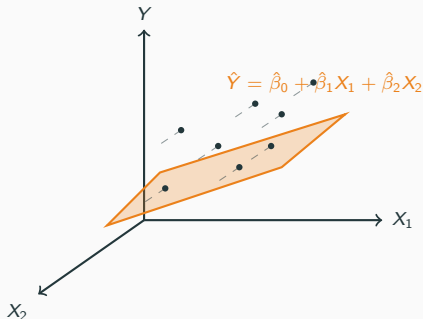
Notation reminder. The multi-predictor vector is

$\mathbf{X} = (X_1, \dots, X_p)^\top \in \mathbb{R}^p$. The intercept β_0 plays the same role as in SLR (the expected Y when all $X_j = 0$).

$p + 1$ unknown coefficients: $\beta_0, \beta_1, \dots, \beta_p$. When $p = 1$ we recover simple linear regression from Lecture 6.

1.3 Geometric Picture

With one predictor, OLS fits a *line* in (X, Y) . With two predictors, OLS fits a *plane* in (X_1, X_2, Y) . With $p \geq 3$ we cannot draw the picture, but the algebra is identical: OLS finds the **hyperplane** closest to the data.



1.4 Interpretation of β_j

Key Result — the conditional reading

β_j is the change in the conditional mean of Y for a one-unit increase in X_j , holding all included predictors fixed.

Two things are new compared to SLR:

- “Holding the other predictors fixed.” The conditional association for X_j is measured after partialling out the others. A variable that looked important in SLR may look useless in multiple regression once correlated predictors are included — and vice versa.
- Each coefficient has its own scientific meaning. β_1 measures TV’s conditional association with sales *after* accounting for radio and newspaper.

We see this in action in §5 (Newspaper drops out once TV and Radio are in the model).

2. OLS in Matrix Form

2.1 Stacking the Data into Matrices

We have n observations of $(Y, X_1, \dots, X_p)$. Observation i gives a scalar equation

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \dots, n.$$

Stack the four ingredients into vectors and a matrix (L8 §8.5 convention — leading column of 1's for the intercept):

$$\underbrace{\mathbf{y}}_{n \times 1} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \underbrace{\mathbf{X}}_{n \times (p+1)} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix},$$
$$\underbrace{\boldsymbol{\beta}}_{(p+1) \times 1} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \underbrace{\boldsymbol{\epsilon}}_{n \times 1} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}.$$

2.2 The Compact Form and the OLS Objective

The n scalar equations collapse to one matrix equation:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

Shape check. $\mathbf{X}\boldsymbol{\beta}$ has shape $(n \times (p + 1)) \cdot ((p + 1) \times 1) = n \times 1$, matching $\mathbf{y}$ and $\boldsymbol{\epsilon}$. ✓

Row check. Row i of $\mathbf{X}\boldsymbol{\beta}$ is $1 \cdot \beta_0 + x_{i1}\beta_1 + \cdots + x_{ip}\beta_p$; adding ϵ_i from $\boldsymbol{\epsilon}$ recovers the scalar equation for y_i . ✓

The OLS objective is the squared length of the residual vector:

$$\text{RSS}(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Goal: find the $\boldsymbol{\beta}$ that minimizes RSS.

2.3 The OLS Estimator — Closed-Form Solution

Minimising $\text{RSS}(\beta)$ over $\beta \in \mathbb{R}^{p+1}$ has a clean closed-form answer.

Key Result — general OLS estimator

Provided $\mathbf{X}^\top \mathbf{X}$ is invertible (the predictors are not exactly linearly dependent),

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

This single formula:

- depends only on the data $(\mathbf{X}, \mathbf{y})$;
- works for any number of predictors p ;
- reduces to the L6 scalar formulas when $p = 1$;
- is what software computes (“computer does it”).

2.3 The OLS Estimator — Closed-Form Solution 2

Where the derivation lives. Obtaining the formula requires two vector-calculus rules and four lines of matrix algebra. We **separate the solution from the derivation** in this lecture: **Appendix A** at the end of these slides contains the rules and the line-by-line derivation, for students who want to see *why* the formula takes this form.

2.4 Compare to Lecture 6

In Lecture 6 we derived OLS for $p = 1$ by:

1. Writing two scalar partial derivatives, $\partial \text{RSS} / \partial \beta_0$ and $\partial \text{RSS} / \partial \beta_1$.
2. Setting both to zero — two normal equations.
3. Solving the 2×2 system by hand.

With $p + 1$ coefficients the same scalar approach would mean writing $p + 1$ partial derivatives and solving a $(p + 1)$ -equation system. The amount of work *grows with* p .

The matrix calculation in Appendix A does not grow. Four lines for any p : expand RSS, take gradient, set to zero, invert. The L6 derivation falls out as the $p = 1$ special case.

This is the payoff for the L8 mini-course: notation now does work that algebra used to do.

3. Inference on Individual Coefficients

3.1 Same Story as Lecture 7

Each $\hat{\beta}_j$ (for $j = 0, 1, \dots, p$) has a standard error $\hat{SE}(\hat{\beta}_j)$ printed in the *Std. error* column of any regression output.

Same convention as Lecture 7: the computer plugs in an estimate of σ and reports $\hat{SE}(\hat{\beta}_j)$. The general formula for all $p + 1$ standard errors involves $(\mathbf{X}^\top \mathbf{X})^{-1}$ and is again the kind of expression we read but do not compute by hand.

Sampling distribution (Key Result, Lecture 7 §3.5 applied per coefficient): for $n \geq 60$,

$$\hat{\beta}_j \stackrel{\text{approx}}{\sim} N(\beta_j, SE(\hat{\beta}_j)^2).$$

3.2 CI and Test — Boxed (L7 Verbatim)

Confidence interval for β_j

$$\hat{\beta}_j \pm z_{\alpha/2} \cdot \widehat{SE}(\hat{\beta}_j)$$

Critical values: 1.645 (90%), 1.96 (95%), 2.576 (99%).

Hypothesis test of $H_0 : \beta_j = 0$

Test statistic

$$\text{stat} = \frac{\hat{\beta}_j}{\widehat{SE}(\hat{\beta}_j)} \stackrel{H_0}{\approx} N(0, 1).$$

Reject if $|\text{stat}| > z_{\alpha/2}$ or $p \leq \alpha$.

Naming aside. Regression software calls this column “t-statistic.” Under our large-sample convention from L4 §3.3, we treat it as approximately $N(0, 1)$: same number, same decision rule.

3.3 The Conditional Reading

The t-test in §3.2 asks

“Is $\beta_j = 0$ holding the other predictors fixed?”

not “Does X_j help predict Y on its own?”

Consequence. A variable can be highly significant in a simple regression (one predictor on its own) and yet fail to reject in a multiple regression (after controlling for others).

Why. If two predictors are correlated, they share the explanatory work. Either one alone looks important; once both are in, each looks marginally less important on top of the other.

Example 1 in §5 illustrates: Newspaper looks fine in a one-at-a-time SLR, but its t-test fails once TV and Radio are also in the model.

4. The F-Test for Joint Significance

4.1 A New Question

Individual t-tests ask, one at a time, “does X_j contribute predictive information, controlling for the others?”

A different question: **does the overall model carry predictive signal?** Or, equivalently: are *all* the slopes zero?

The joint null

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \quad \text{vs} \quad H_1 : \text{at least one } \beta_j \neq 0.$$

(The intercept β_0 is not part of the test — a model with only an intercept is the baseline we’re comparing to.)

4.2 Why Not Just Read the t-Tests?

“If none of the t-tests reject, isn’t the model useless?”

Not quite. Two issues:

- **Multiple comparisons.** With p t-tests at level $\alpha = 0.05$, the chance of *at least one* rejecting by chance alone — even when all β_j truly are zero — is large for moderate p . A naive “one t-test rejects” verdict isn’t a level- α test of the joint null.
- **Joint information.** Several individually weak predictors can be jointly informative.

The **F-test** gives a single, level- α verdict on the whole model.

4.3 The F-Statistic

Key Result — F-statistic

$$F = \frac{(\text{TSS} - \text{RSS}) / p}{\text{RSS} / (n - p - 1)},$$

where

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad \text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Reading the pieces.

- TSS: total variability in Y , ignoring predictors.
- RSS: variability the model leaves unexplained.
- TSS – RSS: variability the model *explains*.

4.4 Intuition

Rewrite the F-statistic as a ratio of averages:

$$F = \frac{\underbrace{\frac{\text{TSS} - \text{RSS}}{p}}_{\text{avg. variance explained per predictor}}}{\underbrace{\frac{\text{RSS}}{n - p - 1}}_{\text{avg. leftover noise per residual df}}}.$$

- ***F* large** (well above 1): the model removes far more variance than what's left in the residuals — the predictors are collectively informative. *Reject H_0 .*
- ***F* near 1**: the model is no better than just fitting the intercept (the mean). *Fail to reject.*

Under H_0 the numerator and denominator have the same expected value, so $F \approx 1$ is what we see when no X_j helps.

4.5 Decision Rule

Under the joint null, F follows an **F-distribution** with degrees of freedom $(p, n - p - 1)$. We don't memorize the distribution's shape and don't use F-tables — the computer reports the p-value directly.

Recipe

1. State $H_0 : \beta_1 = \dots = \beta_p = 0$.
2. Pick α (default 0.05).
3. Read the F-statistic and its p-value from the regression output.
4. **Reject H_0 if $p \leq \alpha$.**
5. Conclude: “at least one predictor is useful” (rejected) or “no evidence any predictor helps” (failed to reject).

Same workflow as the t-tests in §3, just one number for the whole model.

5. Worked Examples

5.1 Example 1: Sales on TV + Radio + Newspaper

Setup. Advertising data, $n = 200$ markets. Regress unit sales (thousands) on three ad-spend channels (thousands of dollars):

$$Y = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{Radio} + \beta_3 \text{Newspaper} + \epsilon.$$

OLS regression output (from the computer):

Predictor	Estimate	Std. error	test stat.	<i>p</i> -value
Intercept	2.939	0.312	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
Radio	0.189	0.0086	21.89	< 0.0001
Newspaper	-0.001	0.0059	-0.18	0.8599

Plus: $F = 570.3$ on $(3, 196)$ df, $p < 0.0001$, $R^2 = 0.897$.

5.1 Example 1: Sales on TV + Radio + Newspaper 2

Individual t-tests ($\alpha = 0.05$):

- **TV.** $|32.81| \gg 1.96$, $p < 0.0001$ — reject. 95% CI: $0.046 \pm 1.96(0.0014) \approx [0.043, 0.049]$, excludes 0. Holding the other channels fixed, each \$1k of TV spend is associated with about 46 more unit sales.
- **Radio.** $|21.89| \gg 1.96$, $p < 0.0001$ — reject. CI $\approx [0.172, 0.206]$.
- **Newspaper.** $|-0.18| < 1.96$, $p = 0.86 > 0.05$ — *fail to reject*. CI: $-0.001 \pm 1.96(0.0059) \approx [-0.013, 0.011]$, contains 0.

F-test. $F = 570 \gg 1$, $p < 0.0001$. **Reject** the joint null — at least one predictor carries predictive signal.

5.1 Example 1: Sales on TV + Radio + Newspaper 3

Putting it together.

- TV and Radio are highly significant; Newspaper is not *once TV and Radio are in the model* — illustrates the conditional reading from §3.3.
- The F-test confirms the overall model is useful; the individual t-tests then identify which predictors are doing the work.
- $R^2 = 0.897$: the three channels together explain about 90% of the variance in sales. (We unpack R^2 in §6.)
- This supports considering a smaller predictive model without Newspaper. Causal budget reallocation would require a stronger design; variable selection is the question of §8.

5.2 Example 2: Starting Salary on GPA + Internship + Major

Setup. $n = 150$ recent graduates of a regional university. Regress starting salary (dollars) on:

- $X_1 = \text{GPA}$ (continuous, 0 to 4).
- $X_2 = \text{Internship}$ (binary: 1 if held a paid internship, 0 otherwise).
- $X_3 = \text{STEM major}$ (binary: 1 if STEM, 0 otherwise).

Two of the three predictors are binary — a preview of §9.

OLS regression output:

Predictor	Estimate	Std. error	test stat.	p -value
Intercept	32,500	4,100	7.93	< 0.0001
GPA	5,800	950	6.11	< 0.0001
Internship	4,200	1,050	4.00	< 0.0001
STEM major	1,800	1,150	1.57	0.119

5.2 Example 2: Starting Salary on GPA + Internship + Major

2

Plus: $F = 28.4$ on $(3, 146)$ df, $p < 0.0001$, $R^2 = 0.37$.

5.2 Example 2: Starting Salary on GPA + Internship + Major

3

Individual t-tests ($\alpha = 0.05$):

- **GPA.** $|6.11| \gg 1.96$, $p < 0.0001$ — reject. 95% CI: $5,800 \pm 1.96(950) \approx [\$3,938, \$7,662]$. Each 1.0-point higher GPA is associated with \$3,938 to \$7,662 more in starting salary, holding internship and major fixed.
- **Internship.** $|4.00| > 1.96$, $p < 0.0001$ — reject. A paid internship is associated with about \$4,200 higher starting salary, holding GPA and major fixed.
- **STEM major.** $|1.57| < 1.96$, $p = 0.119 > 0.05$ — *fail to reject*. The point estimate is \$1,800, but the data don't pin down whether the true coefficient is different from zero. (Caveat: “fail to reject” $\neq$ “ $\beta_3 = 0$ ”; could be sample size.)

5.2 Example 2: Starting Salary on GPA + Internship + Major

4

F-test. $F = 28.4$, $p < 0.0001$ — reject the joint null. The model overall is useful.

5.2 Example 2: Starting Salary on GPA + Internship + Major

5

Putting it together.

- The F-test says the model improves on the grand-mean baseline; the t-tests say GPA and Internship are carrying the predictive signal.
- $R^2 = 0.37$ is much lower than Example 1's 0.897 — these three predictors explain only 37% of salary variability. Lots left over (school, location, negotiation, luck).
- **Both** the F-test and the individual tests are useful: one for the overall verdict, the others for figuring out where the signal is.

6. How Well Does the Model Fit?

6.1 The Variance Decomposition

Definitions — TSS and RSS

- Total sum of squares: $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$ — variability of Y around its mean.
- Residual sum of squares: $RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ — variability the model leaves unexplained.
- Explained variability: $TSS - RSS$.

TSS is fixed (depends only on $\mathbf{y}$). RSS depends on the fitted model: a better-fitting model leaves smaller residuals, hence smaller RSS, hence more explained variance.

6.2 R-Squared

Key Result — coefficient of determination

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} \in [0, 1]$$

Reading. R^2 is the *proportion of Y 's variance explained by the model*.

- $R^2 = 0$: the model is no better than predicting $\bar{y}$ for every observation.
- $R^2 = 1$: the model passes through every data point exactly — perfect fit.

R^2 is reported on every regression output, no work required.

6.3 Reading Our Two Examples

Example 1 (Sales on TV + Radio + Newspaper). $R^2 = 0.897$. The three ad channels together explain about 90% of the variation in sales. Tight model.

Example 2 (Salary on GPA + Internship + Major). $R^2 = 0.37$. The three predictors explain about 37% of salary variation. Lots of unexplained variability — school, location, negotiation, network, luck.

Take-away. R^2 is the single most-quoted goodness-of-fit number. High R^2 does *not* prove a model is causally correct or that predictions on new data will be good. It just says “of the variability in Y *in this sample*, the fitted model accounts for this much.”

6.4 A Caveat — R^2 Can Be Gamed

R^2 never decreases when you add a predictor

Throwing in another X_j — even pure noise — can only leave R^2 the same or push it up. So R^2 alone cannot be used to choose between models with different numbers of predictors.

Why. Adding a predictor gives OLS one more knob to turn to reduce RSS. RSS can only fall (or stay equal). So $R^2 = 1 - \text{RSS}/\text{TSS}$ can only rise.

The fix is the **adjusted R^2** , which penalizes for the number of predictors. Computers report it automatically. Adjusted R^2 becomes important once we start comparing models with different sets of predictors — the next question.

**7. Given X , What Is the
Predicted Y ?**

7.1 Point Prediction

Given a new observation with p predictor values

$\mathbf{x}_{\text{new}} = (x_{\text{new},1}, \dots, x_{\text{new},p})^\top$, form the intercept-augmented vector

$\tilde{\mathbf{x}}_{\text{new}} = (1, x_{\text{new},1}, \dots, x_{\text{new},p})^\top \in \mathbb{R}^{p+1}$. The point prediction is

$$\begin{aligned}\hat{y}_{\text{new}} &= \tilde{\mathbf{x}}_{\text{new}}^\top \hat{\boldsymbol{\beta}} \\ &= \hat{\beta}_0 + \hat{\beta}_1 x_{\text{new},1} + \dots + \hat{\beta}_p x_{\text{new},p}.\end{aligned}$$

Example. For a market with TV = \$100k, Radio = \$20k, Newspaper = \$0, using Example 1's coefficients:

$$\begin{aligned}\hat{y}_{\text{new}} &= 2.939 + 0.046(100) + 0.189(20) + (-0.001)(0) \\ &\approx 11.3 \text{ thousand units.}\end{aligned}$$

A point estimate. But what's the uncertainty around it?

7.2 The Prediction Interval

$\hat{y}_{\text{new}}$ is a point estimate. For a new observation we also want an interval that reflects how much that single observation could plausibly differ from $\hat{y}_{\text{new}}$.

Definition — prediction interval

The **prediction interval** for Y_{new} at predictors $\mathbf{x}_{\text{new}}$ is the interval around $\hat{y}_{\text{new}}$ that contains a single new observation Y_{new} with stated probability (say 95%). It reflects:

- Estimation uncertainty in $\hat{\beta}$ (the regression coefficients are themselves estimates).
- The irreducible noise ϵ (a particular observation will not sit exactly on the population regression hyperplane).

Software reports the prediction interval on request — we don't compute the width by hand (same convention as L7 for the standard error).

7.3 Numerical Example

Continuing the TV = \$100k, Radio = \$20k market from §7.1:

$$\hat{y}_{\text{new}} \approx 11.3 \text{ thousand units (point prediction).}$$

The 95% **prediction interval** for this market's sales (computer-reported) is

$$[7,930, 14,580].$$

Reading. With 95% probability, a particular market with TV = \$100k and Radio = \$20k will have sales somewhere between 7,930 and 14,580 units. The interval is wide because individual markets vary considerably around the population regression hyperplane — even when our coefficient estimates are quite precise.

8. Which Variables Matter?

8.1 The Tempting Idea

Suppose we have p candidate predictors and want to decide which to keep.

Natural approach. For each subset $S \subseteq \{X_1, \dots, X_p\}$, fit a regression using just the predictors in S and score it (adjusted R^2 , AIC, or similar). Pick the subset with the best score.

This is called **best subset selection**. It directly answers “which predictors matter?” — if we can do it.

8.2 The Computational Blow-Up

With p predictors, the number of distinct subsets is 2^p :

p	2^p	comment
10	1,024	easy
20	$\sim 10^6$	slow but possible
30	$\sim 10^9$	hours on a laptop
50	$\sim 10^{15}$	beyond a single machine
100	$\sim 10^{30}$	hopeless

Best subset selection is computationally infeasible once p is moderate. And modern data sets routinely have hundreds or thousands of candidate predictors.

8.3 A Smarter Idea — Selection During the Fit

We need a procedure that picks variables *as part of fitting the model*, instead of fitting 2^p separate models.

Forward pointer to Lecture 10

The **LASSO** solves a modified OLS problem:

$$\min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \sum_{j=1}^p |\beta_j|.$$

The ℓ_1 penalty from L8 §5.2 forces many slope estimates $\hat{\beta}_j$ to come out *exactly zero*, performing variable selection as a side effect of fitting. One model instead of 2^p . Only slopes $j = 1, \dots, p$ are penalized; the intercept β_0 is not.

Lecture 10 unpacks the geometry: why ℓ_1 produces zeros and ℓ_2 does not, why this trades a little bias for a lot of variance reduction, and how to pick the tuning parameter λ .

9. What if X Is Binary?

9.1 Coding 0/1 Indicators

A binary predictor takes only two values: yes/no, treated/control, internship/no internship, owner/renter.

Coding rule. Pick one category to be the baseline (0) and the other to be the indicator (1). The design matrix gets a column of 0s and 1s.

Example (Internship from §5.2):

$$X_2 = \begin{cases} 1 & \text{if held a paid internship} \\ 0 & \text{otherwise.} \end{cases}$$

No new estimator. The OLS formula $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ applies unchanged — some columns of $\mathbf{X}$ are just 0/1 instead of continuous numbers.

9.2 Interpretation

Key Result — binary coefficient

When X_j is a $\{0, 1\}$ indicator, $\hat{\beta}_j$ is the fitted difference in the conditional mean of Y between the two groups, holding all other predictors fixed.

Reading Example 2. Internship coefficient is $\hat{\beta}_2 = 4,200$. Interpretation: *among graduates with the same GPA and the same major*, those who held a paid internship start at salaries about \$4,200 higher on average.

The CI and the t-test from §3 apply to this difference exactly as for continuous predictors — no special machinery.

9.3 More Than Two Categories

A categorical predictor with k levels (e.g., region: Northeast / South / Midwest / West) becomes $k - 1$ indicator columns, with one level chosen as the baseline.

Example. If Northeast is the baseline, the design matrix gets three new columns:

- $X_{\text{South}} = 1$ if South, else 0.
- $X_{\text{Midwest}} = 1$ if Midwest, else 0.
- $X_{\text{West}} = 1$ if West, else 0.

The coefficients are read as *differences from the baseline* (Northeast). Software constructs these columns automatically — we just keep in mind that one level is missing on purpose.

10. Interactions and Higher-Order Terms

10.1 Two Common Extensions

Interaction. Allow the slope relating X_1 to Y to vary with X_2 :

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 X_2) + \epsilon.$$

Treat the product $X_1 X_2$ as a new predictor — its own column in $\mathbf{X}$.

Higher-degree (polynomial). Allow a curve rather than a straight line in X :

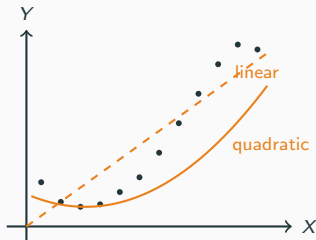
$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon.$$

Treat X^2 and X^3 as new predictors.

10.2 Why OLS Still Works

Even though the model is now *nonlinear in the original predictors*, it is still **linear in the coefficients β_j** .

The OLS formula $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ applies unchanged — just expand $\mathbf{X}$ to include extra columns for products and powers. No new estimator, no new inference machinery. The CIs, t-tests, F-test, and R^2 from §3–§6 all carry over.



Both fits are OLS, on different design matrices: linear uses $(1, X)$; quadratic uses $(1, X, X^2)$.

10.3 The Hard Question We Are Deferring

Once we allow products and powers, candidates multiply quickly:

- p main effects.
- $\binom{p}{2}$ pairwise interactions $X_j X_k$.
- Higher powers $X^2, X^3, \dots$
- Three-way interactions, log transforms, splines, $\dots$

Which to include? With $p = 10$ predictors there are already 45 pairwise interactions. We've reintroduced the variable-selection problem from §8 — worse, because the candidate space is now even larger.

Forward pointer to Lecture 11

Lecture 11 develops **basis expansions**: polynomial and other basis functions become additional columns of the design matrix, producing flexible curves. The degree or number of basis functions controls flexibility.

11. Summary

11.1 Take Away — Foundations

One template, p predictors

- **Model:** $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$, or equivalently $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$.
- **Estimator:** $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ — derived in four lines using the L8 matrix machinery.
- **Inference template** from L7 applies *per coefficient*: each $\hat{\beta}_j$ has a standard error, a CI, and a t-test for $\beta_j = 0$.
- **New:** the F-test gives a single verdict on the whole model.

11.2 Take Away — Key Formulas

The L9 toolbox

- OLS: $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$.
- CI for β_j : $\hat{\beta}_j \pm z_{\alpha/2} \hat{SE}(\hat{\beta}_j)$.
- t-test for $H_0 : \beta_j = 0$: $\text{stat} = \hat{\beta}_j / \hat{SE}(\hat{\beta}_j) \approx N(0, 1)$.
- F-test for joint null: $F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n-p-1)}$, p-value from software.
- Model fit: $R^2 = 1 - \text{RSS}/\text{TSS}$.
- Point prediction: $\hat{y}_{\text{new}} = \tilde{\mathbf{x}}_{\text{new}}^\top \hat{\beta}$; prediction interval for a new observation reported by software.
- Binary X : 0/1 indicators, no new machinery; $\hat{\beta}_j$ reads as group mean difference.
- Interaction / polynomial: expand $\mathbf{X}$ with extra columns; OLS unchanged.

11.3 Where Each Idea Reappears

Forward pointers

- **Lecture 10 (regularization)**. The LASSO and ridge solve the 2^P variable-selection blow-up from §8 by adding a penalty to the OLS objective. The L8 §5 norm ideas become the slope penalties $\sum_{j=1}^P |\beta_j|$ and $\sum_{j=1}^P \beta_j^2$; the intercept is unpenalized. Also: *biased* estimators that can do better than unbiased — breaks the L7 “unbiased is best” intuition.
- **Lecture 11 (non-parametric)**. A systematic basis-expansion framework adds polynomial and other transformed features; the degree or number of basis functions controls the curve’s flexibility.
- **Lecture 12 (logistic regression)**. The same loss-minimization template, but for a binary response $Y \in \{0, 1\}$. Different loss, same workflow.

Appendix A. Deriving the OLS Estimator

A.1 Vector Calculus — Two Rules

To minimise $\text{RSS}(\beta)$ over a vector β , we need to differentiate a function of a *vector* argument. Only two rules, stated as matrix analogues of the scalar versions you already know.

Vector calculus shortcuts

For a vector $\mathbf{a} \in \mathbb{R}^{p+1}$, a symmetric matrix $\mathbf{A} \in \mathbb{R}^{(p+1) \times (p+1)}$, and variable $\beta \in \mathbb{R}^{p+1}$:

$$\nabla_{\beta} (\mathbf{a}^{\top} \beta) = \mathbf{a}, \quad \nabla_{\beta} (\beta^{\top} \mathbf{A} \beta) = 2\mathbf{A}\beta.$$

Compare to scalars. For $\beta \in \mathbb{R}$, $\frac{d}{d\beta}(a\beta) = a$ and $\frac{d}{d\beta}(a\beta^2) = 2a\beta$. The matrix versions look the same, just with $\mathbf{a}$ and $\mathbf{A}$ playing the roles of the constants.

We will use these two rules and nothing else.

A.2 The Derivation

Expand $\text{RSS}(\beta)$:

$$\text{RSS}(\beta) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\beta + \beta^\top (\mathbf{X}^\top \mathbf{X})\beta.$$

Take the gradient using the two rules from A.1:

$$\nabla_{\beta} \text{RSS} = -2\mathbf{X}^\top \mathbf{y} + 2(\mathbf{X}^\top \mathbf{X})\beta.$$

Set to zero — the **normal equations**:

$$\mathbf{X}^\top \mathbf{X}\beta = \mathbf{X}^\top \mathbf{y}.$$

Left-multiply by $(\mathbf{X}^\top \mathbf{X})^{-1}$ (the L8 §7 inverse trick):

A.2 The Derivation 2

Closed-form OLS estimator

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

This is the formula stated in §2.3 of the main body. Four lines of matrix algebra; same answer regardless of p .